Conditional Probability and Bayes Rule

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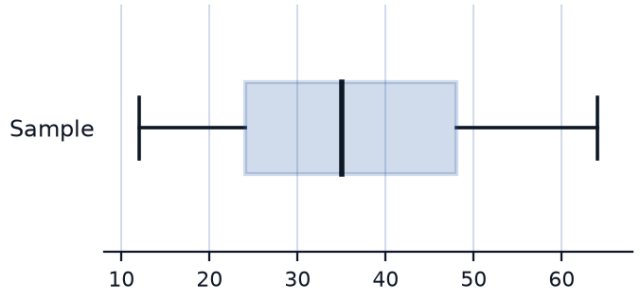
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

[Unit 3 · Probability, Random Variables & Distributions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

2 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A sample dataset has $Q_1 = 48$ and $Q_3 = 76$. Using the 1.5 IQR rule: (a) Determine the exact lower and upper outlier fences. (b) If the dataset includes values of 4 and 122, determine whether each is an outlier.



3 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$.

4 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

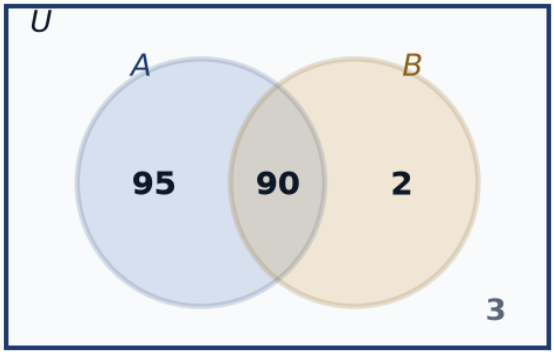
5 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Use Cramer's Rule to solve for x_2 in the linear system: $\{2x_1 + x_2 + x_3 = 4; x_1 - x_2 + 2x_3 = 1; 3x_1 + 2x_2 - x_3 = 7$

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6 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.



7 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 3 · AdvancedAn assembly line fills cereal boxes with a mean weight $\mu = 450$ grams and standard deviation $\sigma = 18$ grams. A random sample of $n = 36$ boxes is selected. What is the mean and standard error of the sample mean weight $\bar{x}$, and what is the probability that $\bar{x}$ exceeds 455 grams?

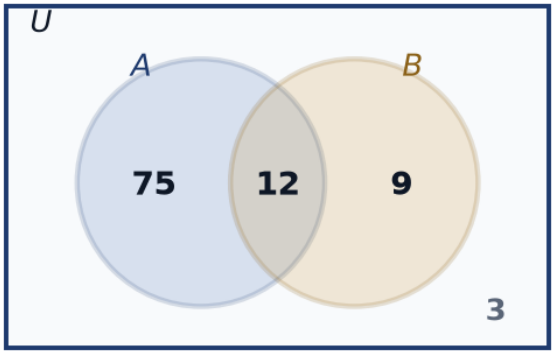
9 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A discrete random variable X has probability distribution: $P(X = 0) = 0.1$, $P(X = 1) = 0.3$, $P(X = 2) = 0.4$, $P(X = 3) = 0.2$. Calculate: (a) $E(X)$, (b) $E(X^2)$, and (c) $\text{Var}(X)$ and σ_X.

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10 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.



11 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

12 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 1 · FoundationsTwo independent events E and F satisfy $P(E) = 0.35$ and $P(F) = 0.60$. Find: (a) $P(E \cap F)$, (b) $P(E \cup F)$, and (c) $P(E \mid F)$.

13 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Level 2 · MediumLet X and Y be independent random variables with $\sigma_X = 6$ and $\sigma_Y = 8$. What is the standard deviation of the difference $D = X - Y$?

14 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

[Unit 2 · Collecting Data, Sampling & Experimental Design · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

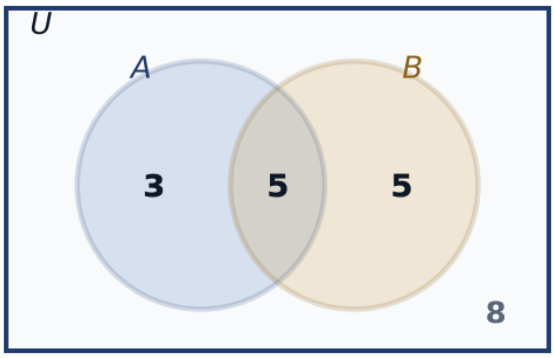
15 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

[Unit 4 · Statistical Inference: Proportions, Means & Slopes · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 6.01 Linear Regression and OLS**

[Linear Transformations, Matrix Representations & Change of Basis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



17 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 7.01 Logistic Regression and Classification**

[Orthogonality, Gram–Schmidt, QR Decomposition & Least Squares · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

18 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[Double Integrals, Fubini's Theorem, Polar Coordinates & The Change-of-Variables Jacobian · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

[Green's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

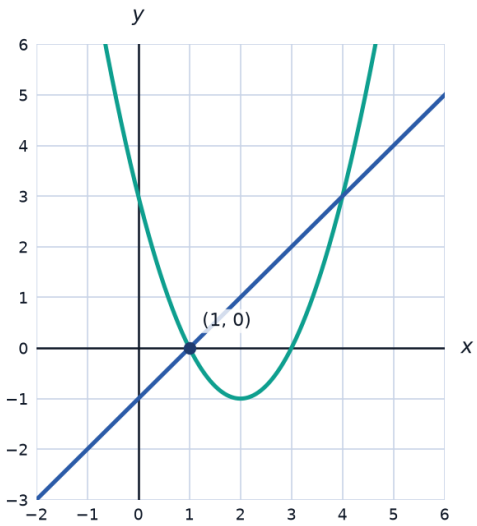
[Functions of Several Variables, Limits, Partial Derivatives & Differentiability · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider the following diagram representing a probability space where events A and B are shown. If $P(A) = 0.6$ and $P(A \cap B) = 0.2$, and the area of the entire sample space is 1, what is the value of $P(B \setminus A)$?



a. 0.2

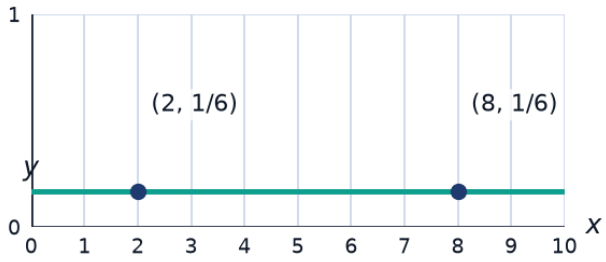
b. 0.4

c. 0.8

d. 0.6

22 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider a continuous random variable X with a uniform distribution over the interval $[2, 8]$. What is the probability that X is between 4 and 7, i.e., $P(4 < X < 7)$?



a. 0.50

b. 0.25

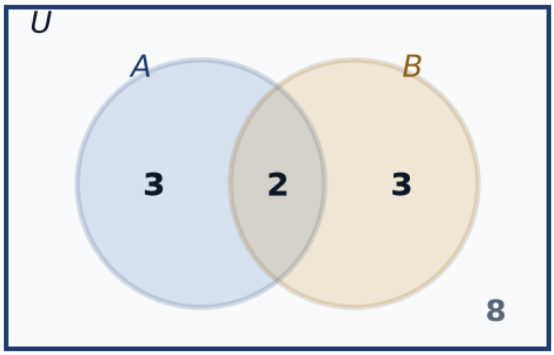
c. 0.75

d. 0.17

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23 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Consider a discrete random variable X that represents the number of heads obtained when a fair coin is tossed 3 times. The probability distribution of X is given by: | x | 0 | 1 | 2 | 3 | |---------|-------|-------|-------|-------| | $P(X = x)$ | $\frac{1}{8}$ | $\frac{3}{8}$ | $\frac{3}{8}$ | $\frac{1}{8}$ | What is the probability that X is less than 2, i.e., $P(X < 2)$?



a. $\frac{3}{8}$

b. $\frac{4}{8}$

c. $\frac{5}{8}$

d. $\frac{6}{8}$

24 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A software company develops a new encryption algorithm. The algorithm's security depends on a sequence of random integers generated by a custom pseudo-random number generator (PRNG). The PRNG is designed such that the probability of generating an integer in a specific range $[a, b]$ is proportional to the length of the range, $b - a$, for $0 \le a \le b \le 1$. Let X be the first integer generated and Y be the second integer generated by this PRNG, independently. The problem is to find the probability that the sum of the first two generated integers, $S = X + Y$, is less than 1. Assume the PRNG generates numbers from a continuous uniform distribution on $[0, 1]$.

a. 1/4

b. 1/3

c. 1/2

d. 2/3

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25 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A quality control engineer is monitoring the number of defects produced by two different machines, Machine A and Machine B. Let X be the number of defects produced by Machine A in a given hour, and Y be the number of defects produced by Machine B in the same hour. The joint probability distribution of X and Y is given in the table below: | $P(X = x, Y = y)$ | $y = 0$ | $y = 1$ | $y = 2$ | | :----------: | :-------: | :-------: | :-------: | | $x = 0$ | 0.10 | 0.15 | 0.05 | | $x = 1$ | 0.20 | 0.30 | 0.10 | | $x = 2$ | 0.05 | 0.05 | 0.00 | What is the expected value of the total number of defects produced by both machines in a given hour, $E(X + Y)$?

a. 1.15

b. 1.30

c. 1.45

d. 1.60

26 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2?

a. 0.16

b. 0.36

c. 1.00

d. 2.50

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27 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A machine dispenses a liquid into containers. The amount of liquid dispensed, in milliliters, follows a continuous uniform distribution over the interval $[95, 105]$. A quality control inspector randomly selects 5 containers and measures the amount of liquid in each. Let X_i be the amount of liquid in the ith container, for $i = 1, 2, 3, 4, 5$. The company has implemented a new calibration process that, on average, increases the mean amount of liquid dispensed by δ milliliters, but also increases the variance by σ^2. The inspector is specifically looking for evidence that the calibration process has *decreased* the variability of the dispensed liquid while maintaining a satisfactory average fill. If the current calibration process yields a mean fill of $\mu_0 = 100$ ml and a standard deviation of $\sigma_0 = 2$ ml, and the new process is hypothesized to have a mean fill of $\mu_1 = 100 + \delta$ ml and a standard deviation of $\sigma_1 = 2 - \sigma_{dev}$ ml, where $\delta > 0$ and $\sigma_{dev} > 0$ are unknown positive constants, what is the minimum value of σ_{dev} such that the probability of falsely concluding that the new process has *increased* variability (i.e., Type I error) is at most 0.01, assuming the mean fill remains at 100 ml (i.e., $\delta = 0$)?

a. The probability cannot be determined without knowing the specific value of δ.

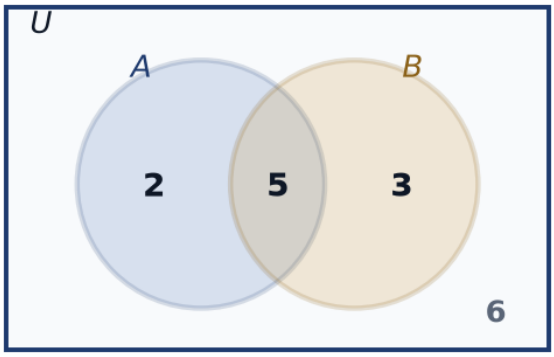
b. Approximately 1.01

c. Approximately 1.37

d. Approximately 2.00

28 [Foundational] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?



a. $\frac{2}{15}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{11}{15}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A discrete random variable X can take on the values 1, 2, 3, and 4. The probability distribution for X is given by P(X=x) = kx, where k is a constant. What is the value of P(X=2)?

a. $\frac{1}{6}$

b. $\frac{1}{3}$

c. $\frac{1}{2}$

d. $\frac{5}{6}$

30 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k?

a. 1/10

b. 1/4

c. 1/6

d. 1/16

31 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. 1/3

c. 1/2

d. 2/3

32 **[Intermediate]** • **Target: Conditional Probability & Bayes' Rule Practice Exercises**

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

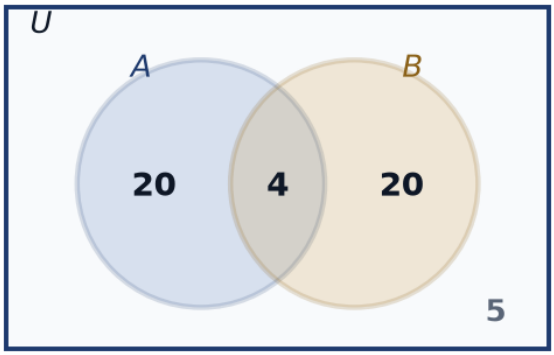
c. $\frac{1}{4}$

d. $\frac{2}{3}$

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33 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

Expected value E(X) for discrete X:



a. The most frequent value

b. $\sum x_i P(x_i)$

c. The median

d. $n \cdot p$ always

34 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

If P(A)=0.4 and P(B|A)=0.5, then P(A∩B)=

a. 0.2

b. 0.5

c. 0.4

d. 0.9

35 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A binomial random variable has $n = 25$ and $p = 0.64$. Find its mean $\mu = np$.

a. 13

b. 14

c. 16

d. 18

36 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

If $P(A) = 0.56$, find $P(A^c)$ (the complement of A).

a. 0.44

b. 0.54

c. 0.64

d. 0.24

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37 [Intermediate] • Target: Conditional Probability & Bayes' Rule Practice Exercises

A geometric random variable has success probability $p = 0.48$. Find the mean (expected number of trials until first success), $\mu = 1/p$.

a. 2.08

b. 8.08

c. 4.08

d. 5.08

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A data set has $Q_1 = 17$ and $Q_3 = 35$. Using the 1.5×IQR rule, which of the following values would be classified as an outlier?

a. 35

b. -14

c. 17

d. -9

39 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Standard deviation of Binomial(n,p) is:

a. np

b. $np(1-p)$

c. $\sqrt{np(1-p)}$

d. $\sqrt{np}$

40 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Probability of event $a \le X \le b$ for PDF f:

a. $\int_a^b f(x)\,dx$

b. $f(b) - f(a)$

c. $f(a) \cdot (b - a)$

d. $F(b) \cdot F(a)$

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41 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Error bound of Simpson's rule on $[a, b]$ with n intervals:

a. $O(h^2)$

b. $O(h^5)$

c. $O(h^3)$

d. $O(h^4)$

42 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Using L'Hôpital's rule: $\lim_{x \to 0} \frac{\sin 3x}{5x} =$

a. 3/5

b. 5/3

c. 0

d. 1

43 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Probability density: $\int_{-\infty}^{\infty} f(x)\,dx =$

a. 0

b. ∞

c. 1

d. 1/2

44 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}} f$ at any point, over all unit vectors $\mathbf{u}$.

a. 23

b. 16

c. 18

d. 17

45 [Challenging] • Target: Conditional Probability & Bayes' Rule Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

For $z = -3xy$ with $x = t^2$, $y = t$, find dz/dt at $t = 3$ using the chain rule.

a. -27

b. 81

c. -81

d. -78

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